

Capacitance and Dielectrics

Storing charge and energy in an electric field: capacitors, dielectrics, defibrillators, touchscreens and the membranes of living cells.

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A **capacitor** is two conductors separated by an insulator. Connect it to a battery and charge $+Q$ collects on one conductor and $-Q$ on the other. Energy is then stored in the electric field between them. Capacitors are everywhere: they smooth power supplies, set the timing of circuits, store the pulse of a camera flash or a defibrillator, sense your finger on a touchscreen, and hold the bits in computer memory. Every cell membrane in your body is a capacitor too. This chapter calculates capacitance for common geometries, combines capacitors, finds the stored energy, and explains how insulating **dielectrics** increase capacitance.

YOU WILL BE ABLE TO

- Define capacitance, and derive it for parallel-plate, cylindrical and spherical capacitors.
- Combine capacitors in series and in parallel, and analyse networks.
- Calculate the energy stored in a capacitor and the energy density of an electric field.
- Explain how dielectrics increase capacitance, and use the dielectric constant and dielectric strength.
- Apply these ideas to defibrillators, cell membranes, sensors and electrostatic actuators.

THINK FIRST

A defibrillator stores about 200 J of energy, which it delivers to the heart in a few milliseconds. A small AA battery stores about 10 000 J. Why do defibrillators use capacitors rather than simply connecting a battery to the patient?

Because of **power**. The defibrillator must deliver its energy very fast: 200 J in about 5 ms is roughly 40 kW, at a high voltage that can drive current through the chest. A battery's chemistry cannot release energy that quickly, and its voltage is only a few volts. The battery is therefore used to *slowly* charge a capacitor, over several seconds, up to more than a thousand volts. The capacitor can then discharge almost instantly (Example 16.4). Capacitors store little energy, but they can release it extremely fast.

§16.1

Capacitance

When a capacitor holds charges $\pm Q$, a potential difference V appears between its conductors, and V is proportional to Q . The constant of proportionality defines the **capacitance**:

$$C = \frac{Q}{V}.$$

The SI unit is the **farad**: $1 \text{ F} = 1 \text{ C/V}$. One farad is enormous. Practical capacitors range from picofarads (pF, 10^{-12} F) through nanofarads and microfarads (μF , 10^{-6} F), up to the thousands of farads of “supercapacitors”.

Capacitance depends only on the **geometry** of the conductors and the **material** between them. It does *not* depend on Q or V .

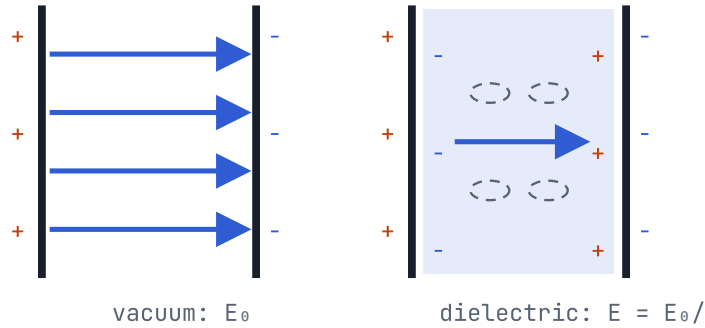
Parallel-plate capacitor

Two plates of area A , a distance d apart, carry charges $\pm Q$. If the plates are large compared with their separation, Gauss's law (Chapter 14) gives a uniform field between them:

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}, \quad V = Ed = \frac{Qd}{\epsilon_0 A}.$$

$$C = \frac{\epsilon_0 A}{d}.$$

A larger area or a smaller gap gives more capacitance.



Left: a parallel-plate capacitor in vacuum, with a uniform field between the plates. Right: a dielectric becomes polarised. The induced surface charges partly cancel the plates' field, so the field inside falls to E_0/κ .

WORKED EXAMPLE 16.1

How big is a farad?

(a) Find the capacitance of two parallel plates, each 2.0 cm^2 , separated by 1.0 mm of air, and the charge on them at 12 V . (b) What plate area, with the same gap, would give 1.0 F ?

Solution

(a) Capacitance and charge:

$$C = \frac{(8.854 \times 10^{-12})(2.0 \times 10^{-4})}{1.0 \times 10^{-3}} = 1.8 \times 10^{-12} \text{ F} = 1.8 \text{ pF}, \quad Q = CV = 21 \text{ pC}.$$

(b) Plate area for 1 F :

$$A = \frac{Cd}{\epsilon_0} = \frac{(1.0)(1.0 \times 10^{-3})}{8.854 \times 10^{-12}} = 1.1 \times 10^8 \text{ m}^2,$$

a square about 11 km on a side.

Evaluate. Supercapacitors reach thousands of farads in a hand-sized package. They use porous carbon electrodes with an enormous internal surface area (around 2000 m^2 per gram), and an electrolyte in which the “gap” is a layer of ions only about a nanometre thick.

Cylindrical and spherical capacitors

Coaxial cylinders (radii $a < b$, length L). From Chapter 15, $V = 2k\lambda \ln(b/a)$ with $\lambda = Q/L$, so

$$C = \frac{2\pi\epsilon_0 L}{\ln(b/a)}.$$

Concentric spheres (radii $a < b$). Here $V = kQ(1/a - 1/b)$, so

$$C = 4\pi\epsilon_0 \frac{ab}{b-a}.$$

Letting $b \rightarrow \infty$ gives the capacitance of an isolated sphere, $C = 4\pi\epsilon_0 R$. Even the whole Earth has only about 710 μF .

WORKED EXAMPLE 16.2

Capacitance of a coaxial cable

Find the capacitance per metre of a coaxial cable with inner radius 0.50 mm and outer radius 3.0 mm, with air between the conductors.

Solution

$$\frac{C}{L} = \frac{2\pi\epsilon_0}{\ln(b/a)} = \frac{2\pi(8.854 \times 10^{-12})}{\ln 6} = 3.1 \times 10^{-11} \text{ F/m} = 31 \text{ pF/m}.$$

Evaluate. A 10 m cable therefore has about 310 pF of capacitance between its conductors. That capacitance must be charged and discharged with every change in the signal, which limits how fast signals can travel along long cables. Engineers specify cables by their capacitance per metre for exactly this reason.

§16.2

Capacitors in combination

KEY IDEA

Parallel (same voltage across each, charges add):

$$C_{\text{eq}} = C_1 + C_2 + \dots$$

Series (same charge on each, voltages add):

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

This is the reverse of the rules for resistors (Chapter 17). Adding capacitors in parallel is like adding plate area. Adding them in series is like increasing the gap.

Why in series the charge is the same. The inner plates of two series capacitors, together with the wire joining them, form an isolated conductor that starts neutral. Whatever charge $+Q$ collects on one inner plate, $-Q$ must collect on the other. So the two capacitors carry the same charge. In series, the **smaller** capacitor takes the **larger** voltage, since $V = Q/C$.

WORKED EXAMPLE 16.3

A capacitor network

$C_1 = 2.0 \mu\text{F}$ and $C_2 = 3.0 \mu\text{F}$ are connected in series. This pair is connected in parallel with $C_3 = 4.0 \mu\text{F}$, and the whole network is connected across a 12 V battery. Find the equivalent capacitance, and the charge on and voltage across each capacitor.

Solution

Series pair:

$$C_{12} = \left(\frac{1}{2.0} + \frac{1}{3.0} \right)^{-1} = 1.2 \mu\text{F}.$$

Total: $C_{\text{eq}} = 1.2 + 4.0 = 5.2 \mu\text{F}$.

The series branch has 12 V across it:

- $Q_1 = Q_2 = 1.2 \mu\text{F} \times 12 \text{ V} = 14.4 \mu\text{C}$.
- $V_1 = 14.4/2.0 = 7.2 \text{ V}$ and $V_2 = 14.4/3.0 = 4.8 \text{ V}$. These add to 12 V. ✓

C_3 also has 12 V across it: $Q_3 = 4.0 \times 12 = 48 \mu\text{C}$.

Check: the total charge from the battery is $14.4 + 48 = 62.4 \mu\text{C}$, and $C_{\text{eq}}V = 5.2 \times 12 = 62.4 \mu\text{C}$. ✓

§16.3

Energy stored in a capacitor

Charging a capacitor takes work. Moving a small charge dq across the potential difference $v = q/C$ needs $dW = (q/C) dq$. Charging it from 0 to Q :

$$U = \int_0^Q \frac{q}{C} dq = \frac{Q^2}{2C} = \frac{1}{2} CV^2 = \frac{1}{2} QV.$$

The factor $\frac{1}{2}$ appears because the voltage builds up gradually during charging. The *average* voltage the charge is moved across is $V/2$.

Where is the energy? It is stored in the **electric field**. For a parallel-plate capacitor:

$$U = \frac{1}{2} CV^2 = \frac{1}{2} \frac{\epsilon_0 A}{d} (Ed)^2 = \frac{1}{2} \epsilon_0 E^2 (Ad).$$

Here Ad is the volume between the plates, so the **energy density** of the field is

$$u = \frac{U}{\text{volume}} = \frac{1}{2} \epsilon_0 E^2.$$

This result holds for any electric field, not just in capacitors.

WORKED EXAMPLE 16.4

A defibrillator

A defibrillator charges a $150 \mu\text{F}$ capacitor to 1.6 kV. (a) How much energy does it store? (b) It discharges through the patient's chest, of resistance about 50Ω . The discharge time scale is RC (Chapter 17). Estimate the duration of the pulse and the average power delivered.

Solution

(a) Stored energy:

$$U = \frac{1}{2} CV^2 = \frac{1}{2} (150 \times 10^{-6}) (1600)^2 = 192 \text{ J}.$$

(b) Time scale:

$$\tau = RC = 50(150 \times 10^{-6}) = 7.5 \text{ ms.}$$

Average power over this time:

$$P \sim \frac{192 \text{ J}}{7.5 \text{ ms}} \approx 26 \text{ kW.}$$

The peak current is $V/R = 1600/50 = 32 \text{ A}$.

Evaluate. The capacitor turns a gentle, seconds-long charging current from a battery into a brief, intense pulse. That pulse depolarises the whole heart muscle at once, so that the heart's natural pacemaker can take over again. Modern defibrillators shape the pulse into two phases ("biphasic"), which works at lower energies (120–200 J) with less damage to tissue.

WORKED EXAMPLE 16.5

Why capacitors store so little energy

Find the largest possible energy density in an air-filled capacitor, given that air breaks down at $3.0 \times 10^6 \text{ V/m}$. Compare it with the chemical energy density of petrol, about $3.4 \times 10^{10} \text{ J/m}^3$.

Solution

$$u_{\text{max}} = \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} (8.854 \times 10^{-12}) (3.0 \times 10^6)^2 = 40 \text{ J/m}^3.$$

That is a billion times less than petrol.

Evaluate. Even with the best dielectrics, which allow fields of about 10^9 V/m and have κ around 10, capacitors store far less energy per volume than batteries or fuels. They are chosen when speed of delivery, or millions of charge cycles, matter more than total energy: camera flashes, defibrillators, regenerative braking, and smoothing power supplies.

§16.4

Dielectrics

Filling the space between the plates with an insulator, a **dielectric**, increases the capacitance by a factor κ called the **dielectric constant**:

$$C = \kappa C_0, \quad \text{for example} \quad C = \frac{\kappa \epsilon_0 A}{d}.$$

Why it works. The field polarises the dielectric's molecules. Polar molecules partly line up with the field, and non-polar molecules have dipoles induced in them. Inside the material the dipoles cancel, but at the two surfaces next to the plates they leave thin layers of **bound charge**, opposite in sign to the charge on the adjacent plate (see the figure above). These bound charges partly cancel the plates' field:

$$E = \frac{E_0}{\kappa}.$$

For a given charge on the plates, the voltage drops by the factor κ , so $C = Q/V$ rises by κ . Equivalently, for a given voltage, the plates can hold κ times as much charge.

Wherever ϵ_0 appears in the electrostatics of a dielectric medium, it is replaced by the **permittivity** $\epsilon = \kappa \epsilon_0$. For example, Coulomb's law in water is weaker by a factor of $\kappa = 80$. This is why salts dissolve in water: the attraction between Na^+ and Cl^- is cut 80-fold.

MATERIAL	κ	DIELECTRIC STRENGTH (MV/M)
Vacuum	1 (exactly)	—
Air (1 atm)	1.0006	3
Paper	3.7	16
Polyethylene	2.3	20–50
Glass	5–10	10
Mica	5.4	100–200
Water (20 °C)	80	— (it conducts)

MATERIAL	κ	DIELECTRIC STRENGTH (MV/M)
Cell membrane (lipid)	about 5–8	—
Barium titanate ceramic	1000–10 000	1–10

The **dielectric strength** is the largest field a material can withstand before it breaks down and conducts, usually with a spark that damages it. It sets a capacitor's maximum working voltage.

WORKED EXAMPLE 16.6

Inserting a dielectric, with and without the battery

A 10 pF air capacitor is charged to 100 V. A slab of dielectric with $\kappa = 4.0$ is then slid in to fill the gap completely. Find the new charge, voltage and stored energy if (a) the battery was disconnected before the slab was inserted, and (b) the battery stays connected.

Solution

Initially: $Q_0 = CV = 1.0 \text{ nC}$ and $U_0 = \frac{1}{2}CV^2 = 50 \text{ nJ}$. The new capacitance is $C = 40 \text{ pF}$.

(a) **Isolated**, so Q is fixed at 1.0 nC:

$$V = \frac{Q}{C} = 25 \text{ V}, \quad U = \frac{Q^2}{2C} = 12.5 \text{ nJ}.$$

The energy *falls*. The missing 37.5 nJ is the work done by the field in **pulling the slab in**: the fringing field at the edges of the plates attracts the slab's polarised charges.

(b) **Connected**, so V is fixed at 100 V:

$$Q = CV = 4.0 \text{ nC}, \quad U = \frac{1}{2}CV^2 = 200 \text{ nJ}.$$

The battery moves an extra 3.0 nC through 100 V, supplying 300 nJ. Of this, 150 nJ raises the stored energy, and the other 150 nJ is the work done pulling the slab in.

Lesson. Always ask what is held fixed: the charge (isolated capacitor) or the voltage (connected to a battery).

WORKED EXAMPLE 16.7

The cell membrane as a capacitor

Biological membranes have a specific capacitance of about $1.0 \text{ } \mu\text{F}/\text{cm}^2$. (a) Taking the membrane thickness as 7.0 nm, find its effective dielectric constant. (b) For a spherical cell

of radius $10\text{ }\mu\text{m}$ with a membrane potential of 70 mV , find the membrane capacitance and the number of excess monovalent ions needed on each side.

Solution

(a) $1.0\text{ }\mu\text{F}/\text{cm}^2 = 0.010\text{ F}/\text{m}^2$. Since $C/A = \kappa\epsilon_0/d$:

$$\kappa = \frac{(C/A)d}{\epsilon_0} = \frac{0.010(7.0 \times 10^{-9})}{8.854 \times 10^{-12}} = 7.9.$$

That is typical of a lipid layer.

(b) The membrane area is $4\pi(10 \times 10^{-6})^2 = 1.26 \times 10^{-9}\text{ m}^2$, so

$$C = 0.010(1.26 \times 10^{-9}) = 1.3 \times 10^{-11}\text{ F} = 13\text{ pF}.$$

$$Q = CV = (1.3 \times 10^{-11})(0.070) = 8.8 \times 10^{-13}\text{ C} \Rightarrow N = \frac{Q}{e} \approx 5.5 \times 10^6\text{ ions}.$$

Evaluate. That sounds like a lot, but the cell contains around 10^{11} – 10^{12} potassium ions, so the charge imbalance is only about one part in 10^5 . Inside and outside, the fluids remain electrically neutral to an excellent approximation. Only a thin sheet of charge lines each face of the membrane, exactly like the charges on a capacitor's plates. Problem P16.12 explores this further.

IN PRACTICE

Capacitors as sensors. Because $C = \kappa\epsilon_0 A/d$, anything that changes the area, the gap or the dielectric changes the capacitance, and electronics can measure changes as small as a femtofarad (10^{-15} F).

- **Touchscreens.** A grid of transparent electrodes forms tiny capacitors. A finger, which is conductive and full of water, distorts their fields and changes their capacitance where it touches.
- **MEMS accelerometers** in phones and airbag controllers detect the motion of a proof mass as a change in the gap between interdigitated “comb” electrodes (Problem P16.17).
- **Condenser microphones** sense the vibration of a thin diaphragm that forms one plate of a capacitor.
- **Capacitive humidity sensors** respond to water entering a polymer dielectric, since water’s κ is large.
- **Computer memory.** Each bit of DRAM is a tiny capacitor, holding about 25 fF and refreshed thousands of times a second.

CHAPTER SUMMARY

- $C = Q/V$, in farads. Parallel plates: $C = \kappa\epsilon_0 A/d$. Coaxial cylinders: $2\pi\kappa\epsilon_0 L / \ln(b/a)$. Spheres: $4\pi\kappa\epsilon_0 ab/(b-a)$. Isolated sphere: $4\pi\epsilon_0 R$.
- Parallel: $C_{\text{eq}} = \sum C_i$ (same V). Series: $1/C_{\text{eq}} = \sum 1/C_i$ (same Q).
- Energy: $U = Q^2/2C = \frac{1}{2}CV^2 = \frac{1}{2}QV$. Field energy density: $u = \frac{1}{2}\epsilon_0 E^2$, or $\frac{1}{2}\kappa\epsilon_0 E^2$ in a dielectric.
- A dielectric polarises: $E = E_0/\kappa$ and $C = \kappa C_0$. The dielectric strength limits the field and so the working voltage.
- Isolated capacitor: Q is fixed. Capacitor connected to a battery: V is fixed.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P16.1

A capacitor's charge is doubled. What happens to its capacitance, its voltage and its stored energy?

P16.2

Explain, in terms of molecular polarisation, why filling a capacitor with a dielectric increases its capacitance.

P16.3

A $1\ \mu\text{F}$ and a $10\ \mu\text{F}$ capacitor are connected in series across a battery. Which has the larger charge? Which has the larger voltage across it?

P16.4

An isolated, charged parallel-plate capacitor has its plates pulled farther apart. What happens to the charge, the field between the plates, the voltage and the stored energy? Where does any extra energy come from?

LEVEL B – STANDARD PROBLEMS

P16.5

A capacitor has square plates $5.0\ \text{cm}$ on a side, separated by a $0.20\ \text{mm}$ sheet of mica ($\kappa = 5.4$, dielectric strength $100\ \text{MV/m}$). Find its capacitance and its maximum safe voltage.

P16.6

Three identical $6.0\ \mu\text{F}$ capacitors are available. Find every distinct equivalent capacitance that can be made using all three.

P16.7

$C_1 = 4.0\ \mu\text{F}$ is connected in series with a parallel pair, $C_2 = 2.0\ \mu\text{F}$ and $C_3 = 6.0\ \mu\text{F}$. The combination is connected to $24\ \text{V}$. Find the equivalent capacitance, and the charge on and voltage across each capacitor.

P16.8

A camera flash charges a $1000\ \mu\text{F}$ capacitor to $400\ \text{V}$, then discharges it in about $1.0\ \text{ms}$. Find the stored energy and the average power of the flash.

P16.9

(a) Find the capacitance of two concentric spherical shells of radii $10\ \text{cm}$ and $12\ \text{cm}$. (b) Find the capacitance of the Earth, treated as an isolated conducting sphere of radius $6370\ \text{km}$.

P16.10

A $10\ \mu\text{F}$ capacitor charged to $100\ \text{V}$ is disconnected from the battery, then connected across an identical uncharged capacitor. Find the final voltage, the charge on each capacitor, and the total stored energy before and after. Where has the “missing” energy gone?

P16.11

Defibrillator design. A defibrillator must store $200\ \text{J}$, with its capacitor charged to $2.0\ \text{kV}$. (a) What capacitance is needed? (b) What charge is stored? (c) If the energy is delivered in about $10\ \text{ms}$, what is the average current?

P16.12

Membrane charge. A neuron’s cell body is roughly a sphere of radius $15\ \mu\text{m}$. Its membrane has a specific capacitance of $1.0\ \mu\text{F}/\text{cm}^2$, and the resting potential is $70\ \text{mV}$. (a) Find the membrane capacitance and the charge separated across it. (b) How many potassium ions does this represent? (c) The cytoplasm contains $140\ \text{mM}$ of potassium. How many potassium ions are in the cell, and what fraction must cross the membrane to set up the resting potential?

LEVEL C – CHALLENGE PROBLEMS**P16.13**

A parallel-plate capacitor (area A , gap d) contains a dielectric slab of thickness $t < d$ and dielectric constant κ , lying parallel to the plates. Show that $C = \frac{\epsilon_0 A}{d - t + t/\kappa}$. Check the limits $t = 0$, $t = d$ and $\kappa \rightarrow \infty$ (a metal slab).

P16.14

A parallel-plate capacitor (area A , gap d) is half filled with a dielectric of constant κ , so that the dielectric covers half the area and spans the full gap. Find the capacitance. Compare it with filling half the gap across the whole area (Problem P16.13 with $t = d/2$). Which arrangement gives more capacitance?

P16.15

Electrostatic force. (a) For an isolated parallel-plate capacitor carrying charge Q , use $F = -dU/dx$ at constant Q to show that the plates attract with a force $F = Q^2/(2\epsilon_0 A) = \frac{1}{2}QE$. (b) Explain why the force is $\frac{1}{2}QE$ rather than QE . (c) A MEMS electrostatic actuator has plates of area 1.0 mm^2 , $2.0 \text{ }\mu\text{m}$ apart, at 10 V . Find the force.

P16.16

Field energy of a charged sphere. (a) By integrating the energy density $\frac{1}{2}\epsilon_0 E^2$ over all space outside a conducting sphere of radius R and charge Q , show that the total field energy is $kQ^2/(2R)$, equal to $\frac{1}{2}QV$. (b) Suppose the electron's rest energy $m_e c^2$ were entirely the energy of its electric field, with its charge spread over the surface of a sphere. Estimate the radius, and compare it with the "classical electron radius" $ke^2/(m_e c^2) = 2.82 \text{ fm}$.

P16.17

MEMS accelerometer. A proof mass forms the middle plate of two capacitors, each of capacitance 1.00 pF when the mass is centred, with a $2.00 \text{ }\mu\text{m}$ gap on each side. (a) An acceleration displaces the mass by 10 nm towards one side. Find the change in each capacitance. (b) Explain why measuring the *difference* between the two capacitances is better than measuring one. (c) If the mass of $1.0 \text{ }\mu\text{g}$ is held by a suspension of stiffness 0.50 N/m , what acceleration produces the 10 nm displacement?

P16.18

A dielectric slab (κ) of width w (equal to the width of the plates) is partly inserted, to a distance x , into a parallel-plate capacitor of gap d that is kept connected to a battery of voltage V . (a) Show that the capacitance is $C(x) = \frac{\epsilon_0 w}{d} [L + (\kappa - 1)x]$, where L is the plate length. (b) Show that the force pulling the slab in is $F = \frac{(\kappa - 1)\epsilon_0 w V^2}{2d}$, independent of x . (Careful: at constant V , the battery also does work, and $F = +\frac{dU}{dx}$.) (c) Evaluate F for $w = 10$ cm, $d = 1.0$ mm, $V = 1000$ V and $\kappa = 3.0$.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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